

The Goldilocks of Pragmatic Understanding: Fine-Tuning Strategy Matters for Implicature Resolution by LLMs

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Abstract

Despite widespread use of LLMs as conversational agents, evaluations of performance fail to capture a crucial aspect of communication: interpreting language *in context*—incorporating its pragmatics. Humans interpret language using beliefs and prior knowledge about the world. For example, we intuitively understand the response “I wore gloves” to the question “Did you leave fingerprints?” as meaning “No”. To investigate whether LLMs have the ability to make this type of inference, known as an *implicature*, we design a simple task and evaluate four categories of widely used state-of-the-art models. We find that, despite only evaluating on utterances that require a binary inference (yes or no), models in three of these categories perform close to random. However, LLMs instruction-tuned at the example-level perform significantly better. These results suggest that certain fine-tuning strategies are far better at inducing pragmatic understanding in models. We present our findings as the starting point for further research into evaluating how LLMs interpret language in context and to drive the development of more pragmatic and useful models of human discourse.

1 Introduction

User: “Have you seen my phone?”
 GPT-3: “Yes, I have seen your phone.”

GPT-3’s response² is a perfectly fine answer to the question, but a human might answer differently. They might respond “it’s in your bag,” bypassing the obvious follow-up question (“where is it?”). Giving such a helpful and efficient answer is an example of pragmatic language use that goes beyond the mere production of semantically plausible and consistent utterances. Meaning is not only determined by a combination of words, but also context, beliefs, and social institutions (Wittgenstein, 1953; Grice, 1975; Huang, 2017). Consider another exchange where Esther asks her friend Juan “Can you come to my party on Friday?” and Juan responds “I have to work”. We resolve Juan’s response as him declining the invitation by using the contextual commonsense knowledge that having to work on a Friday night precludes attendance. Both these exchanges contain an *implicature*—utterances that convey something other than their literal meaning.³ Implicatures illustrate how context contributes to meaning; distinguishing writing and speaking from communicating (Green, 1996). We cannot fully

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²Appendix D contains details on how this completion was obtained from text-davinci-002

³In Appendix E we present an introduction to implicature.

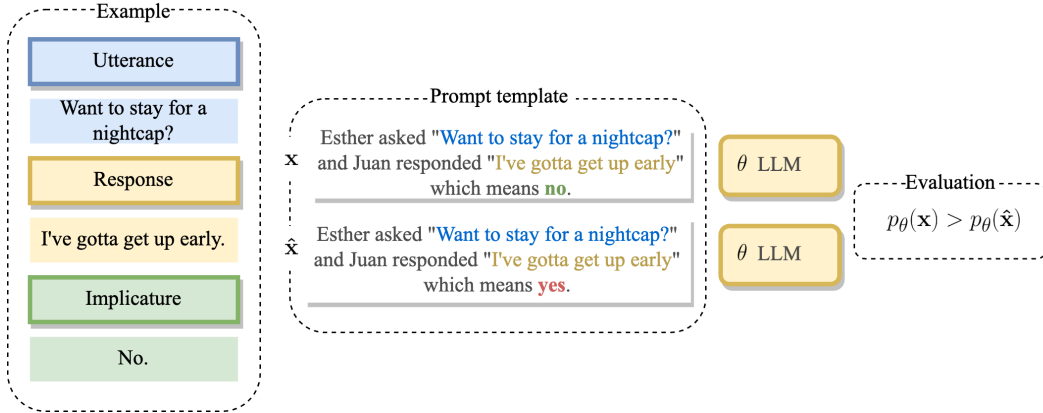


Figure 1: A schematic depiction of the protocol we propose to evaluate whether language models can resolve implicatures. Each example in the test set gets wrapped in templates and transformed into an *incoherent* example by swapping “yes” and “no”. The model is said to resolve the implicature if it assigns a higher likelihood to the coherent text than the incoherent text.

understand utterances without understanding their implications. Indeed, the term “communication” presupposes the speaker’s implications are understood by the addressee. Being able to resolve completely novel implicatures and, more broadly, engage in pragmatic understanding constitutes an essential and ubiquitous aspect of our every day use of language.

Large language models (LLMs) have demonstrated remarkable ability on a variety of downstream tasks such as planning (Huang et al., 2022), commonsense reasoning (Kojima et al., 2022), information retrieval (Lewis et al., 2020; Kim et al., 2022) and code completion (Austin et al., 2021; Biderman and Raff, 2022), to name a few. When fine-tuned with human feedback, LLMs obtain higher ratings on desiderata like helpfulness (Ouyang et al., 2022; Bai et al., 2022), and are proposed as conversational agents (Thoppilan et al., 2022). Despite the widespread use of LLMs as conversational agents, there has been limited evaluation of their ability to navigate contextual commonsense knowledge.

This raises an important question: *to what extent can large language models resolve conversational implicature?* To answer this question we use a public dataset of conversational implicatures and propose an evaluation protocol on top of it (Figure 1). We evaluate a range of state-of-the-art models that can be categorised into four groups; large-scale pre-trained models, like OPT (Zhang et al., 2022), LLMs fine-tuned on conversational data, like BlenderBot (Ng et al., 2019), LLMs fine-tuned on common NLP benchmarks with natural instructions for each benchmark, like Flan-T5 (Chung et al., 2022), and LLMs fine-tuned on tasks with natural instructions for each example, e.g. versions of OpenAI’s InstructGPT-3 series⁴. Our results show that implicature resolution is a challenging task for LLMs. All pre-trained models obtain close to random zero-shot accuracy (around 60%), whereas humans obtain 86%. However, our results suggest that instruction-tuning at the example level is important for pragmatic understanding. Models fine-tuned with this method perform much better than others, and analysis of different model sizes shows that they have the best scaling properties. We further push performance for these models with chain-of-thought prompting, and find that one model in the group (GPT-4) reaches human-level performance. In summary, we conclude that pragmatic understanding has not yet arisen from large-scale pre-training *on its own*, but scaling analysis shows that it might for much larger scale. Fine-tuning on conversational data or benchmark-level instructions does not produce models with pragmatic understanding. However, fine-tuning on instructions at the example-level is a fruitful path towards more useful models of human discourse.

The **main contributions** of this work are: i) we motivate implicature understanding as a crucial aspect of communication that is currently mostly missing from evaluations of LLMs, ii) we design an implicature resolution task and propose a comprehensive evaluation protocol on which we evaluate both humans and LLMs to find that it poses a significant challenge for SotA LLMs, and iii) we provide a thorough analysis of the results and identify one fine-tuning strategy (instruction-tuning at the example-level) as a promising method that produces models with more pragmatic understanding.

⁴The precise method is unpublished and differs from the original instructGPT (Ouyang et al., 2022).

2 Related Work

LLMs have demonstrated remarkable performance on tasks for which they were not explicitly trained (Brown et al., 2020). Building on the hypothesis that these abilities arise due to implicit multitask learning (Radford et al., 2019), the recent works of Sanh et al. (2022) and Wei et al. (2022) explicitly train LLMs in a supervised multitask fashion, leading to models that are better zero-shot learners with fewer parameters. Besides rapidly saturating language understanding benchmarks (Kielbaso et al., 2021), these advancements make LLMs beneficial foundations for agents performing a plethora of tasks (Adolphs et al., 2022; Reed et al., 2022). The trend towards using these models as agents brings along with it increased urgency for alignment with human values (Kenton et al., 2021). However, larger models trained with next-word prediction are generally more toxic and unhelpful (Gehrmann et al., 2020; Bender et al., 2021; Lin et al., 2022). Recent work mitigates this with methods like prompting and finetuning on human-annotated outputs (Askell et al., 2021; Ouyang et al., 2022; Thoppilan et al., 2022). The produced models are more aligned on desiderata such as informativeness when evaluated by dedicated benchmarks and humans. We argue, however, that there is still something missing in these benchmarks. What is helpful and informative, as Kasirzadeh and Gabriel (2022) also point out, depends on the context in which a conversation is held. Consequently, any application that requires communicating with humans will rely on pragmatic communication skills—something that is not explicitly captured by the benchmarks used to evaluate the alignment of LLMs.

There is a large body of work that investigates the interplay between pragmatics and computational modeling (Cianflone et al., 2018; Schuster et al., 2020; Louis et al., 2020; Kim et al., 2021; Li et al., 2021; Jeretic et al., 2020; Parrish et al., 2021; Hosseini et al., 2023). Cianflone et al. (2018) introduce the task of predicting adverbial presupposition triggers, which are words like ‘again’ that trigger the unspoken presupposition that an event has happened before. Schuster et al. (2020) study the ability of computational models to do scalar inferences, finding that models use linguistic features to make pragmatic inferences. Kim et al. (2021) find that a substantial part of question-answering datasets contain questions that are unanswerable due to false presuppositions (i.e. “which linguist invented the lightbulb”). Hosseini et al. (2023) present a dataset for selecting entities with indirect answers, and find that language models adapted for this task get reasonable accuracy, but that there is room for improvement. The difference with this body of work and ours is that we look at the emergence of pragmatic understanding from large-scale language modeling. Jeretic et al. (2020); Parrish et al. (2021) are early works investigating the emergence of pragmatic understanding in pretrained language models, but they only look at scalar implicatures and presuppositions. Zheng et al. (2021) are the first to evaluate pretrained language models on conversational implicatures. This is important pioneering work highlighting the difficulty of implicature for language models, but their evaluations require task-specific training and the models they evaluate are relatively small. In contrast, our evaluation protocol is applicable out-of-the-box and is much more comprehensive, evaluating models up to 176 billion parameters and using in-context prompting. Additionally, Zheng et al. (2021) benchmark synthetic data whereas this work evaluates performance on naturally occurring implicatures (George and Mamidi, 2020). We believe this to be a better representation of the true distribution of implicatures in natural dialogue.

The standard set of benchmarks LLMs are evaluated on covers many tasks, but even though implicature is one of the most important aspects of language pragmatics (Levinson, 1983), it is only evaluated as part of BIG-bench (Srivastava et al., 2022). Unfortunately, the methodology used by the BIG-bench implicature task contributors has limitations, which call into question the validity of their claims. Firstly, the task contributors discard a subset of the data that is ambiguous according to them. In our view this defeats the point of the benchmark. Implicatures are a type of non-literal, ambiguous language the intended meaning of which humans often easily interpret; comparing the way humans and models do this is precisely what we are interested in. In turn, we expect performance on the BIG-bench task to overestimate the ability of LLMs to resolve naturally occurring implicatures. We keep this challenging subset of the data and instead use human evaluation to deal with examples that are too ambiguous to understand. Secondly, the difference in performance between their average and best rater is 18%, whereas for our evaluations this difference is 6%. This indicates their human evaluation is of low quality, but it is impossible to verify because there are no details available on how the annotation is done. Finally, BIG-bench uses only base LLMs and no SotA fine-tuning methods. In summary, we use a more challenging dataset, and in turn at least six times more evaluations per model, we provide higher-quality human annotations, and evaluate four different categories of LLMs to investigate which aspects of LLMs contribute to their performance on implicature understanding.